

Question 1 — Simple Calculator (HTML + CSS + JS)

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Simple Calculator</title>

  <link rel="stylesheet" href="style.css">
</head>

<body>

  <div class="calculator">

    <h1>Simple Calculator</h1>

    <input type="number" id="num1" placeholder="Enter first number">

    <input type="number" id="num2" placeholder="Enter second number">

    <div class="buttons">
      <button onclick="add()">Add</button>
      <button onclick="subtract()">Subtract</button>
      <button onclick="multiply()">Multiply</button>
      <button onclick="divide()">Divide</button>
    </div>

    <p id="result">Result</p>

  </div>

  <script src="script.js"></script>

</body>
</html>
```

Style.css

```
body {
  font-family: Arial, sans-serif;
  background-color: lightgray;
```

```
display: flex;
justify-content: center;
align-items: center;

height: 100vh;
}

.calculator {
  background-color: white;
  width: 350px;

  padding: 25px;

  text-align: center;

  border: 1px solid gray;
  border-radius: 10px;
}

h1 {
  margin-bottom: 20px;
}

input {
  width: 90%;

  padding: 10px;
  margin: 8px;

  border: 1px solid gray;
  border-radius: 5px;
}

button {
  padding: 10px 15px;
  margin: 5px;

  border: none;
  border-radius: 5px;

  cursor: pointer;
}

button:hover {
```

```
background-color: lightgray;
}

#result {
  margin-top: 20px;

  font-size: 18px;
  font-weight: bold;
}
```

Script.js

```
function add() {

  let num1 = Number(document.getElementById("num1").value);
  let num2 = Number(document.getElementById("num2").value);

  let result = num1 + num2;

  document.getElementById("result").innerText = "Result: " + result;
}

function subtract() {

  let num1 = Number(document.getElementById("num1").value);
  let num2 = Number(document.getElementById("num2").value);

  let result = num1 - num2;

  document.getElementById("result").innerText = "Result: " + result;
}

function multiply() {

  let num1 = Number(document.getElementById("num1").value);
  let num2 = Number(document.getElementById("num2").value);

  let result = num1 * num2;

  document.getElementById("result").innerText = "Result: " + result;
}

function divide() {
```

```

let num1 = Number(document.getElementById("num1").value);
let num2 = Number(document.getElementById("num2").value);

if (num2 === 0) {

    document.getElementById("result").innerText =
        "Cannot divide by zero";

} else {

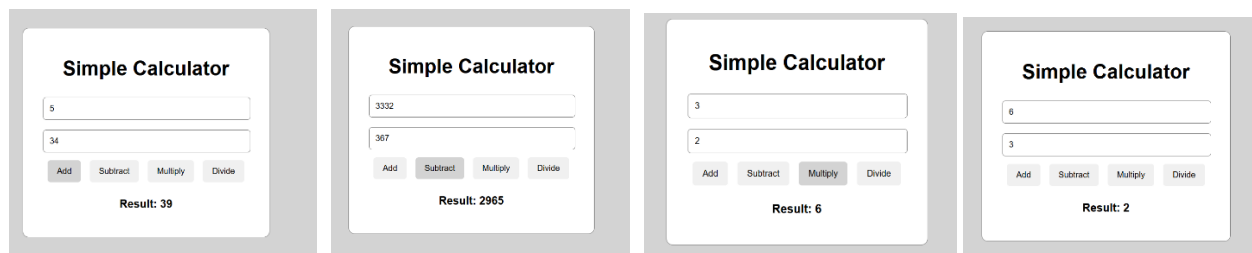
    let result = num1 / num2;

    document.getElementById("result").innerText =
        "Result: " + result;

}
}

```

Output:



Question 2 — Functions With and Without Parameters

Script.js

```

// Function without parameters
function greet() {
    console.log("Welcome to JavaScript!");
}

// Function with one parameter
function greetUser(name) {
    console.log("Welcome, " + name + "!");
}

// Function with two parameters
function addNumbers(num1, num2) {
    return num1 + num2;
}

```

```
// Calling greet()
greet();

// Calling greetUser()
greetUser("Ali");

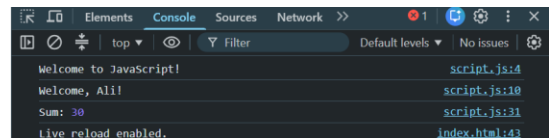
// Calling addNumbers()
let sum = addNumbers(10, 20);

console.log("Sum:", sum);
```

output:

Q2 - Functions

Open the browser Console to see the output.



Question 3 — Number Pattern Using Loops

Index.js

```
<!DOCTYPE html>
<html>
<head>
  <title>Q3 - Number Pattern</title>
</head>

<body>

  <h1>Number Pattern</h1>

  <p>Open the Console to see the pattern.</p>

  <script src="script.js"></script>

</body>
</html>
```

Script.js

```
// Ask the user to enter a number
let n = parseInt(prompt("Enter a number:"));

// Outer loop for rows
for (let i = 1; i <= n; i++) {
```

```

let row = "";

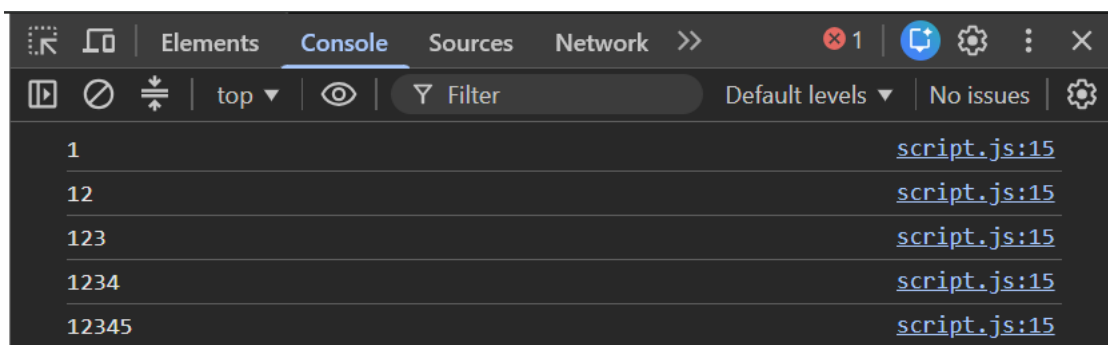
// Inner loop to print numbers
for (let j = 1; j <= i; j++) {

    row = row + j;
}

console.log(row);
}

```

Output:



Question 4 — Scenario: Roll Call Using break and continue

```

<!DOCTYPE html>
<html>
<head>
    <title>Q3 - Number Pattern</title>
</head>

<body>

    <h1>Number Pattern</h1>

    <p>Open the Console to see the pattern.</p>

    <script src="script.js"></script>

</body>
</html>

```

```

// Ask the user to enter a number
let n = parseInt(prompt("Enter a number:"));

// Outer loop for rows

```

```
for (let i = 1; i <= n; i++) {  
  
    let row = "";  
  
    // Inner loop to print numbers  
    for (let j = 1; j <= i; j++) {  
  
        row = row + j;  
    }  
  
    console.log(row);  
}
```

Output:

```
Calling roll number 1  
Calling roll number 2  
Calling roll number 3  
Calling roll number 4  
Calling roll number 5  
Calling roll number 6  
Calling roll number 7  
Calling roll number 8  
Calling roll number 9  
Calling roll number 10  
Calling roll number 11  
Calling roll number 12  
Calling roll number 14  
Calling roll number 15  
Calling roll number 16  
Calling roll number 17  
Total roll numbers called = 16
```